

Menus



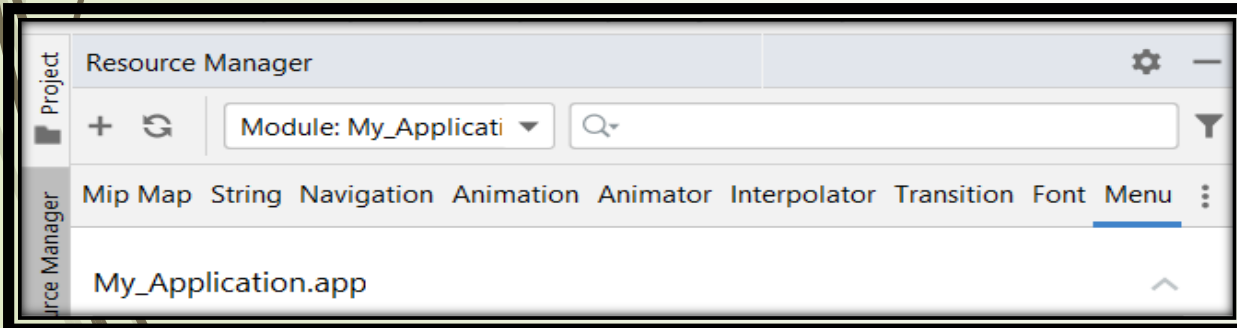
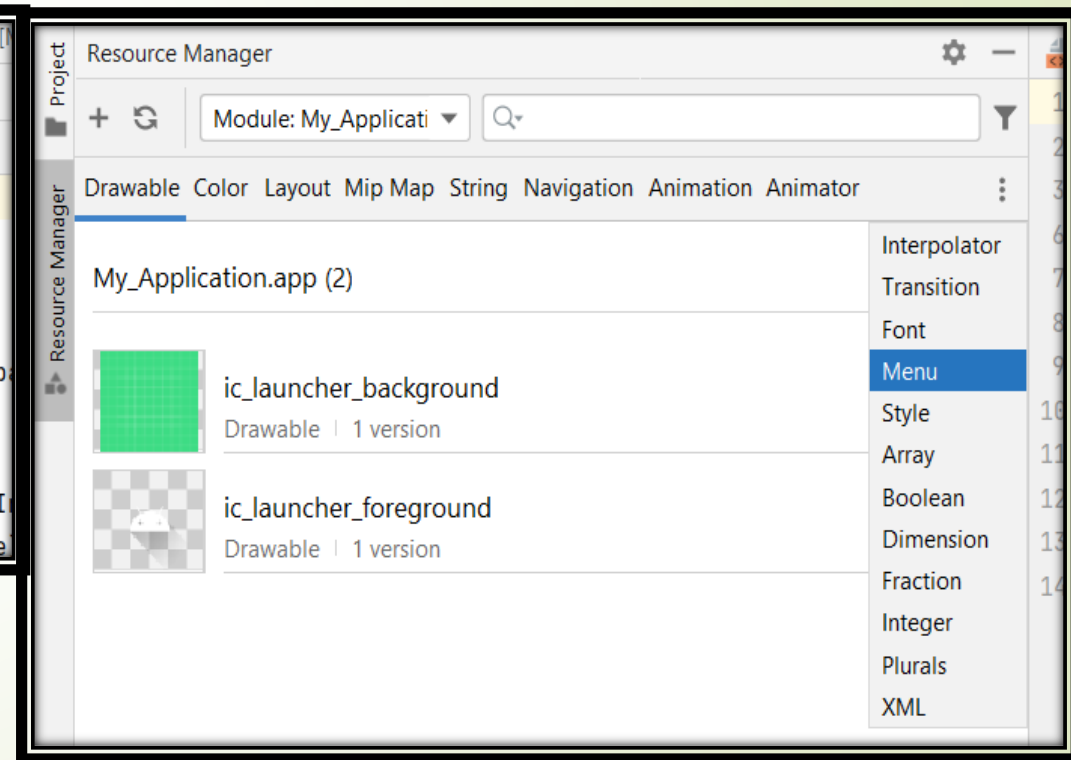
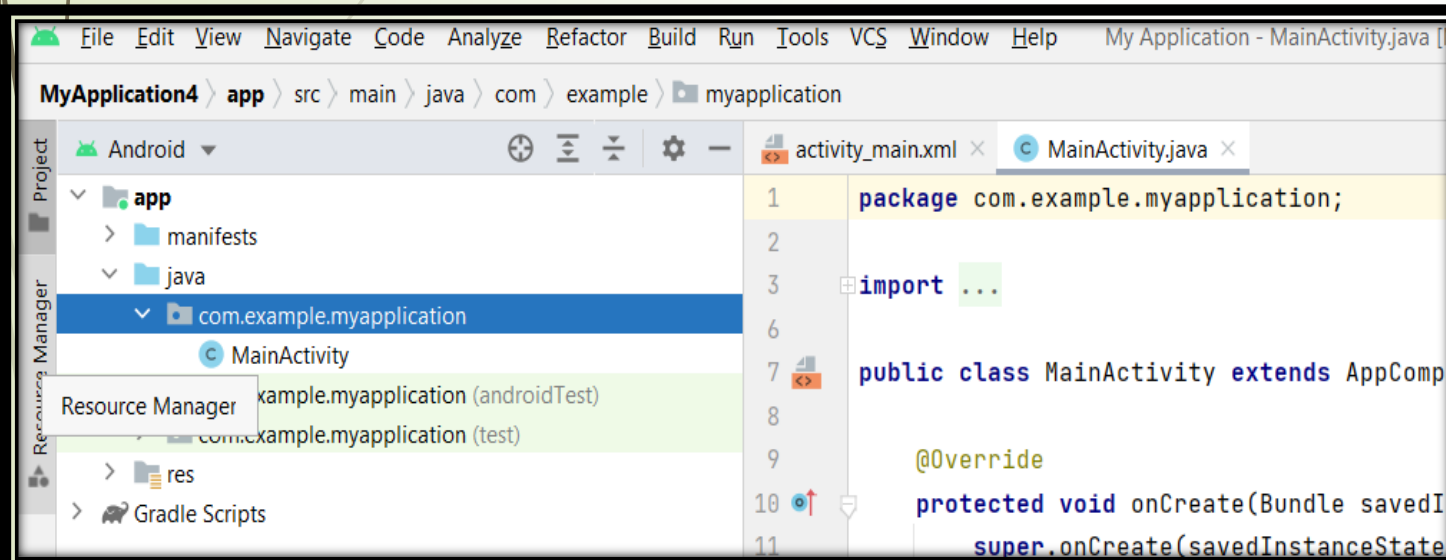
Menus

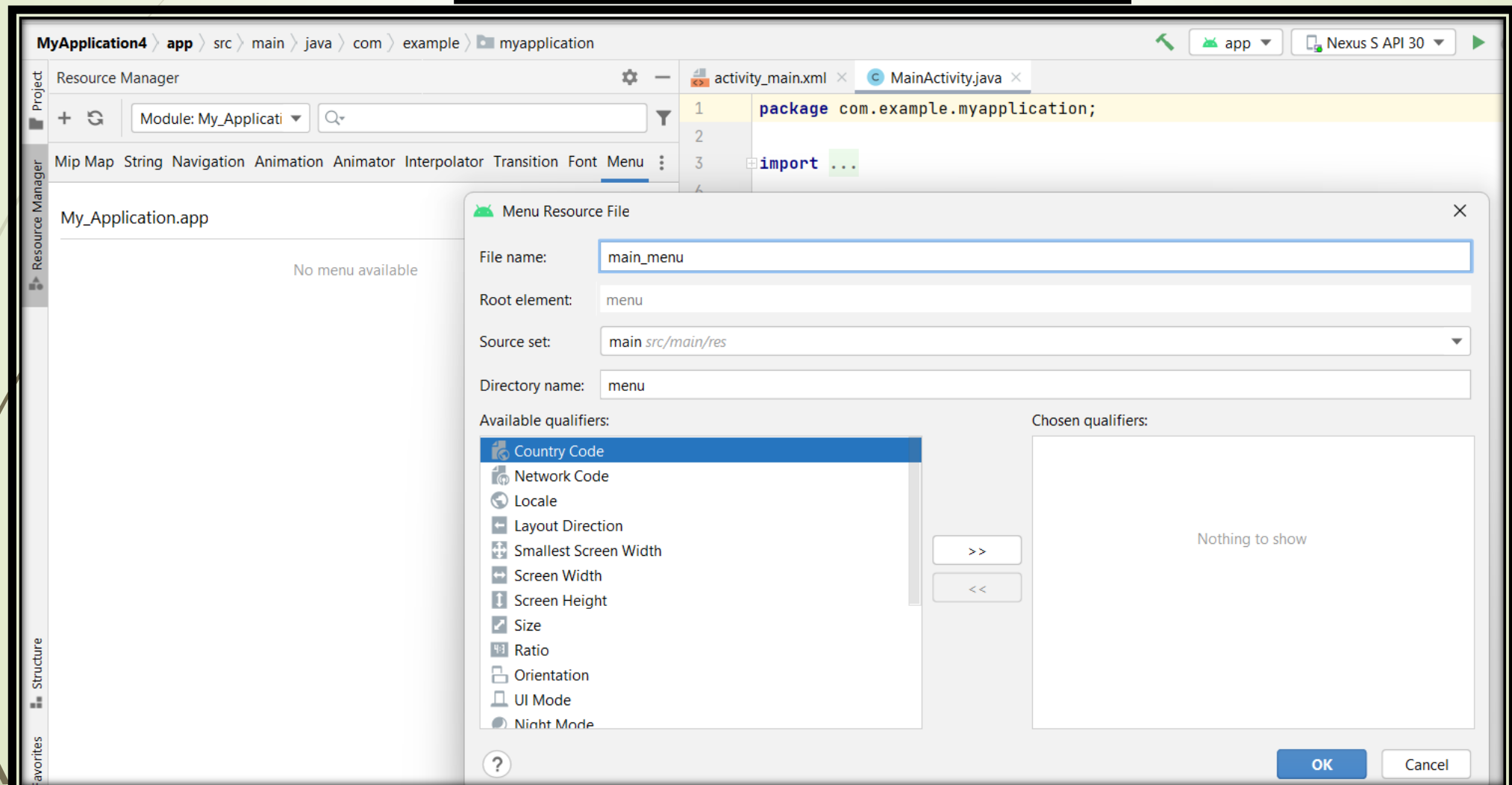
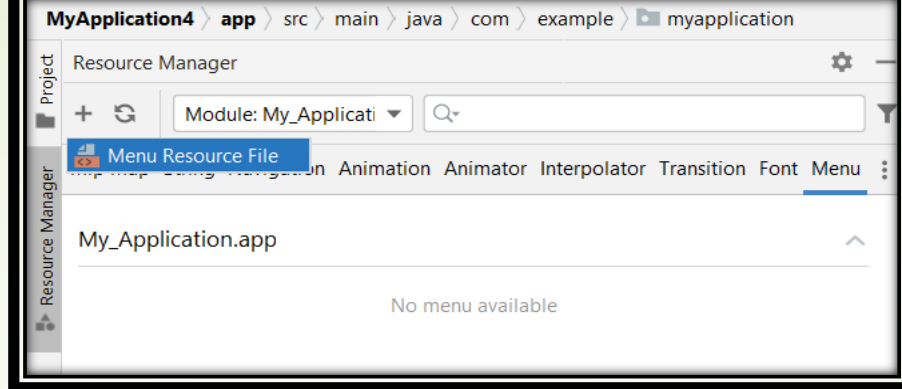
Three types of the menus –

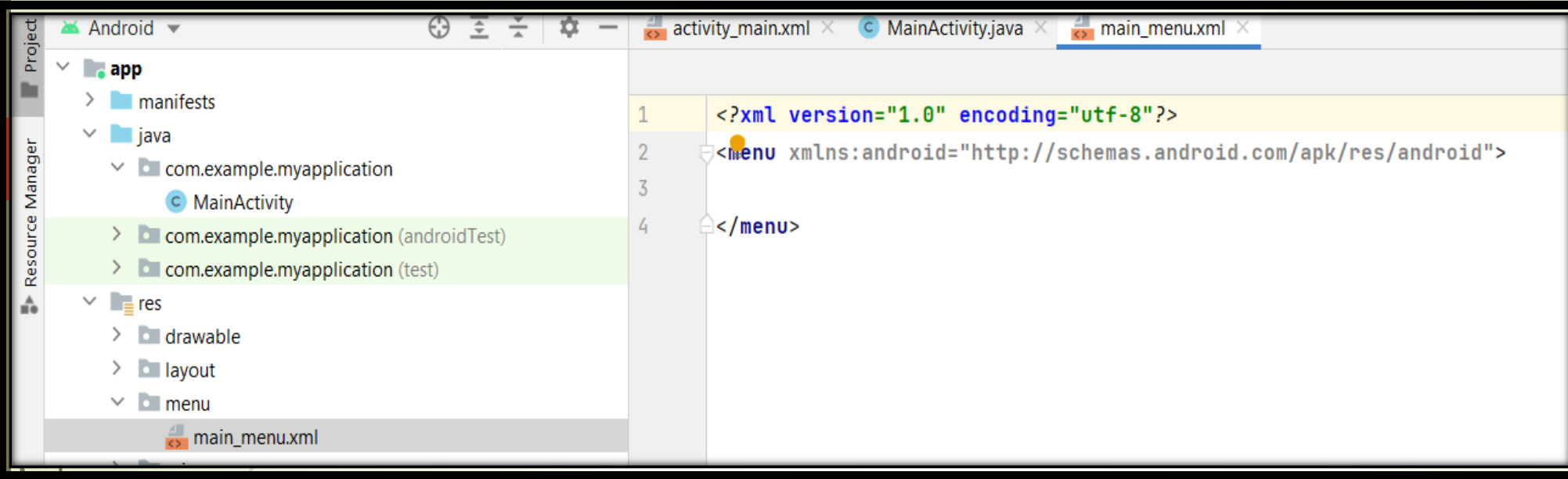
- Option Menu
- Context Menu
- Pop-up Menu

How to Add Menu File

Go to Resource Manager → click on triple Dot → select menu →
Click on + → select add Menu Resource File → Give name and click
ok







main_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/android"
        android:title="Android" />
    <item
        android:id="@+id/java"
        android:title="JAVA"/>
    <item
        android:id="@+id/kotlin"
        android:title="Kotlin"/>
</menu>
```

Option Menus

- Android Option Menus are the primary menus of android.
- They can be used for settings, search, delete item etc.
- Here, we are inflating the menu by calling the inflate() method of MenuInflater class.
- To perform event handling on menu items, you need to override onOptionsItemSelected() method of Activity class.

Activity_main.xml

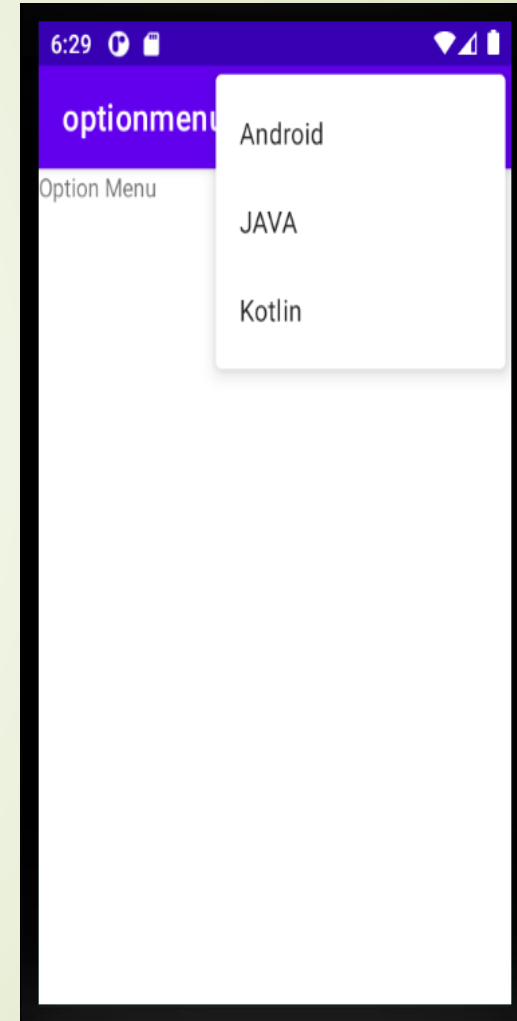
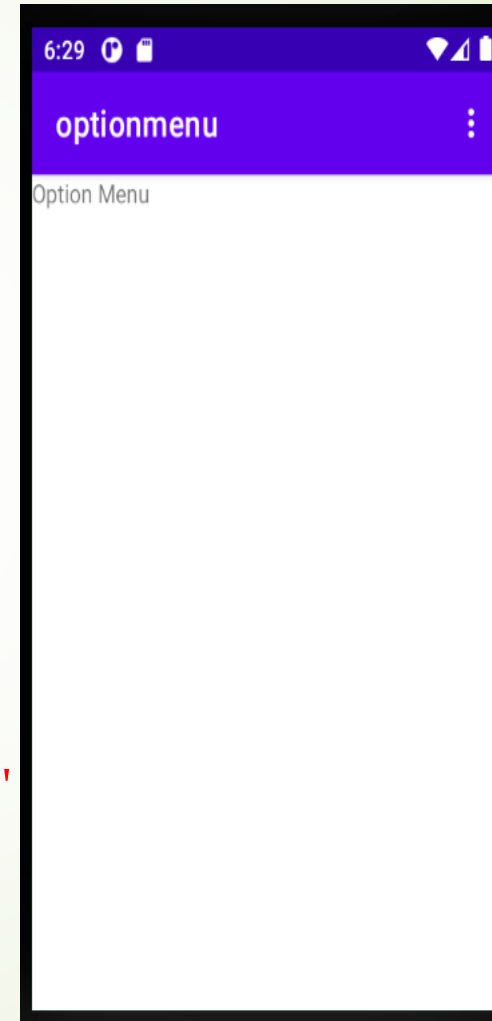
```
<?xml version="1.0" encoding="utf-8"?>
```

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/a"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

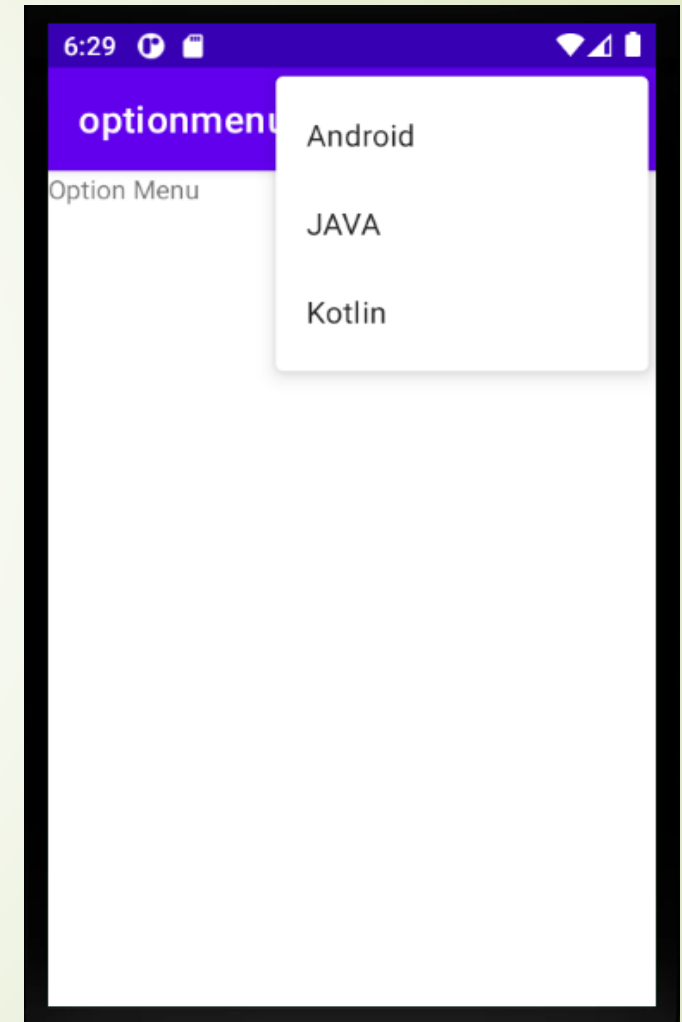
```
<TextView
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Option Menu"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
```

```
</RelativeLayout>
```

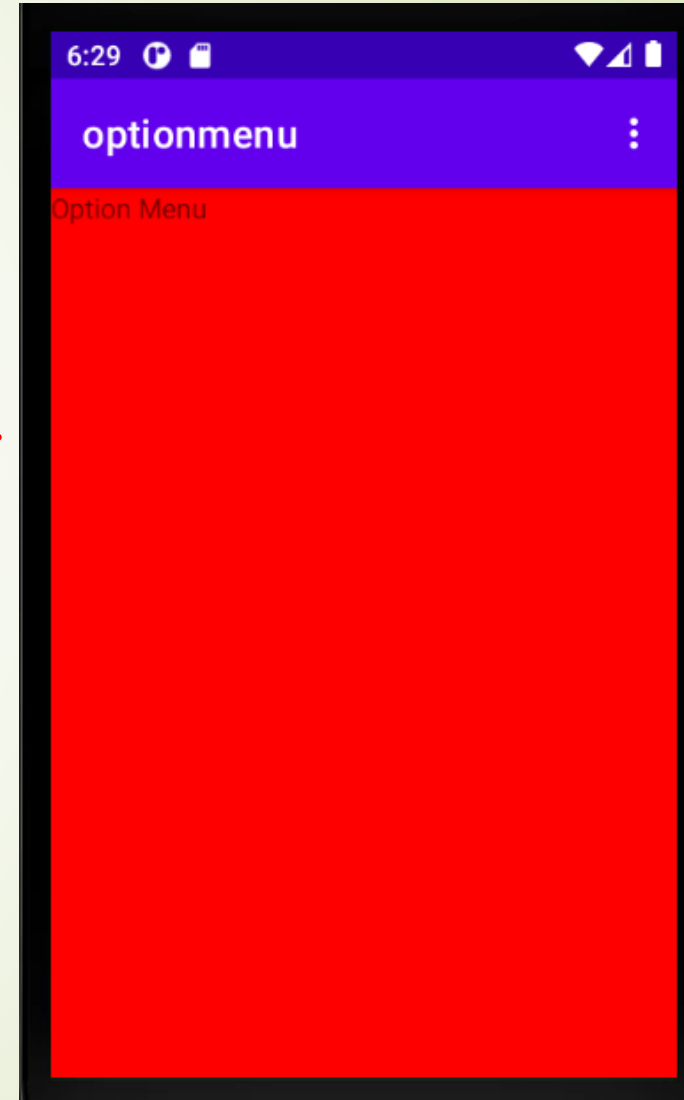



```
package com.example.optionmenu;
import android.graphics.Color;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.RelativeLayout;
public class MainActivity extends AppCompatActivity {
    RelativeLayout currentLayout;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        currentLayout =(RelativeLayout) findViewById(R.id.a);
    }
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.main_menu, menu);
        return true;
    }
}
```



@Override

```
public boolean onOptionsItemSelected(MenuItem item) {  
    switch (item.getItemId()) {  
        case R.id.android:  
            currentLayout.setBackgroundColor(Color.RED);  
            return false;  
        case R.id.java:  
            currentLayout.setBackgroundColor(Color.GREEN);  
            return false;  
        case R.id.kotlin:  
            currentLayout.setBackgroundColor(Color.BLUE);  
            return false;  
        default:  
            return super.onOptionsItemSelected(item);  
    }  
}
```



Context Menu

- Android context menu appears when user **press long click on the element.**
- It is also known as floating menu.
- It affects the selected content while doing action on it.
- It doesn't support item shortcuts and icons.

Activity_main.xml

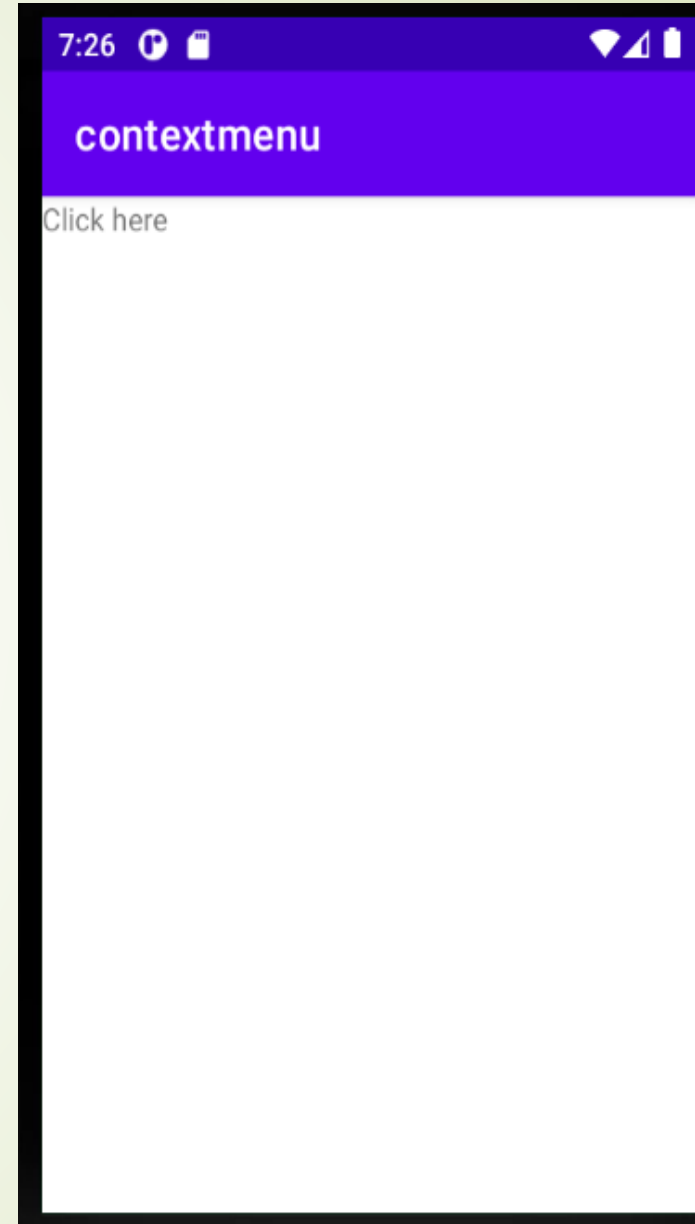
```
<?xml version="1.0" encoding="utf-8"?>
```

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/rel"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
<TextView
```

```
    android:layout_width="wrap_content"
    android:id="@+id/text"
    android:layout_height="wrap_content"
    android:text="Click here"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
```

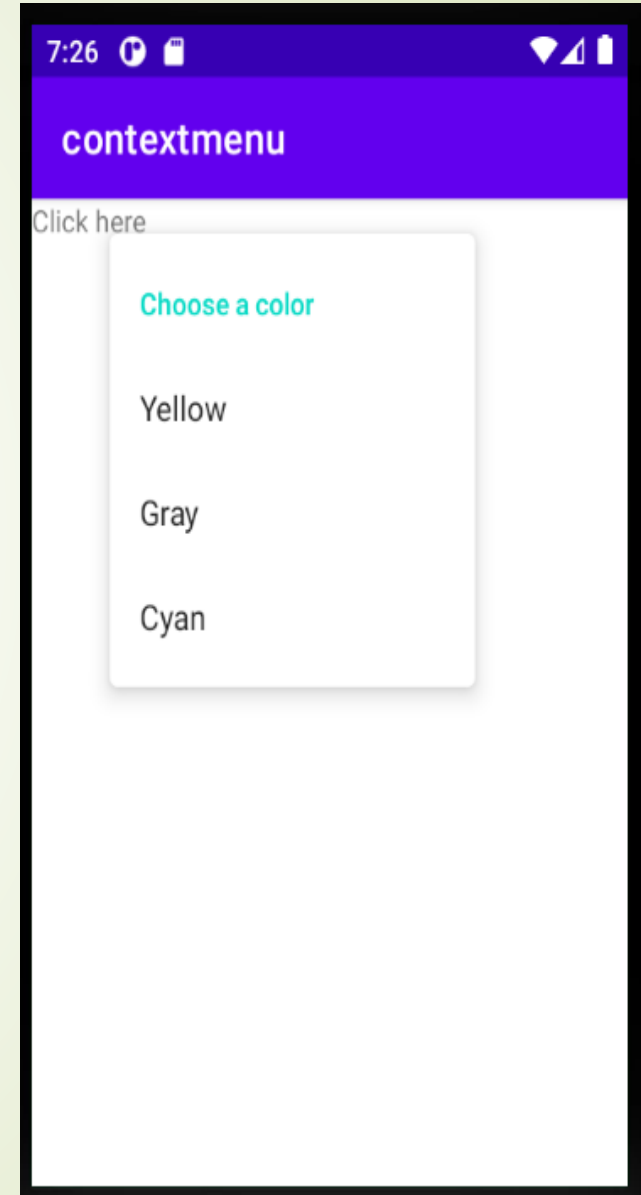
```
</RelativeLayout>
```



```
package com.example.contextmenu;
import android.graphics.Color;
import android.view.ContextMenu;
import android.view.MenuItem;
import android.widget.RelativeLayout;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity {
    TextView textView;
    RelativeLayout r;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        textView = (TextView) findViewById(R.id.text);
        r = (RelativeLayout) findViewById(R.id.rel);
        // here you have to register a view for context menu you can register any view
        // like listview, image view, textview, button etc
        registerForContextMenu(textView);
    }
}
```

@Override

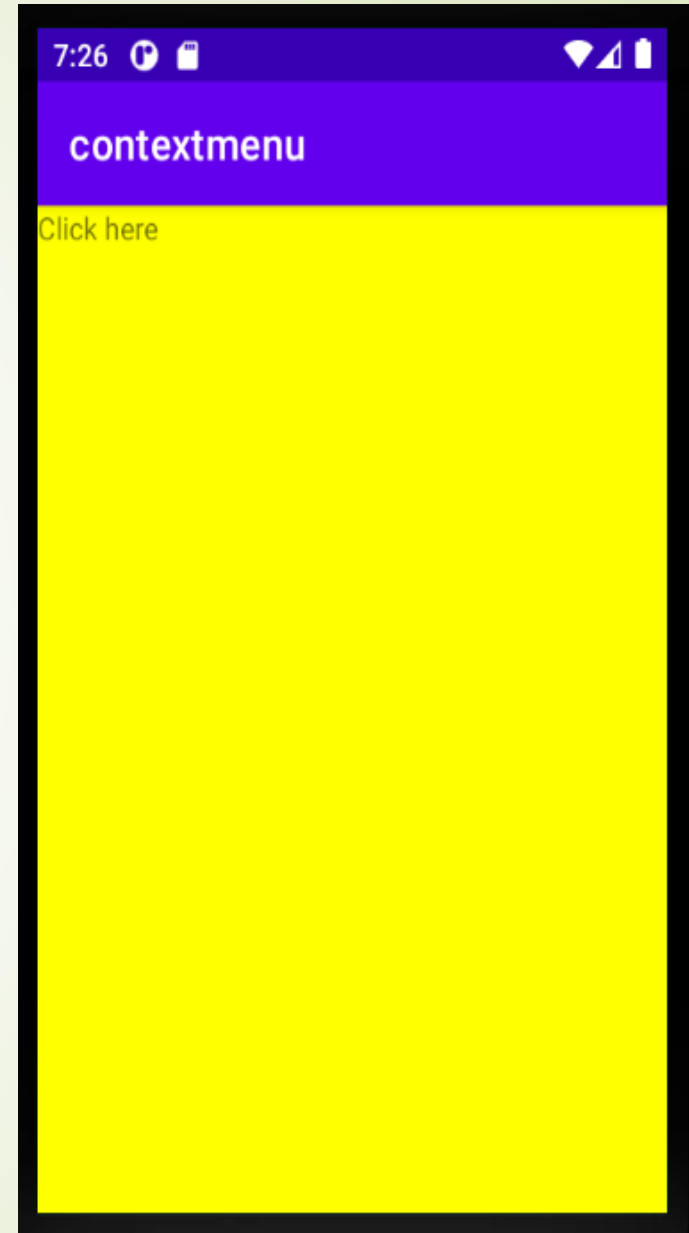
```
public void onCreateContextMenu(ContextMenu menu, View v,  
                                ContextMenu.ContextMenuInfo menuInfo) {  
    super.onCreateContextMenu(menu, v, menuInfo);  
    // you can set menu header with title icon etc  
    menu.setHeaderTitle("Choose a color");  
    // add menu items  
    menu.add(0, v.getId(), 0, "Yellow");  
    menu.add(0, v.getId(), 0, "Gray");  
    menu.add(0, v.getId(), 0, "Cyan");  
}
```



// menu item select listener

@Override

```
public boolean onOptionsItemSelected(MenuItem item) {  
    if (item.getTitle() == "Yellow") {  
        r.setBackgroundColor(Color.YELLOW);  
    } else if (item.getTitle() == "Gray") {  
        r.setBackgroundColor(Color.GRAY);  
    } else if (item.getTitle() == "Cyan") {  
        r.setBackgroundColor(Color.CYAN);  
    }  
    return true;  
}  
}
```



Pop-up Menu

- Android Popup Menu displays the menu below the anchor text if space is available otherwise above the anchor text.
- It disappears if you click outside the popup menu.
- The `android.widget.PopupMenu` is the direct subclass of `java.lang.Object` class.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

<Button

```
    android:id="@+id/clickBtn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:background="#0F9D58"
    android:text="Click Me"
    android:textColor="#ffffff"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintLeft_toLeftOf="parent"
    app:layout_constraintRight_toRightOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```



package com.example.popmenu;

MainActivity.java

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

import android.view.MenuItem;

import android.view.View;

import android.widget.Button;

import android.widget.PopupMenu;

import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

@Override

protected void onCreate(Bundle savedInstanceState) {

super.onCreate(savedInstanceState);

setContentView(R.layout.activity_main);

Button button = (Button) findViewById(R.id.clickBtn);

// Setting onClick behavior to the button

```
button.setOnClickListener(new View.OnClickListener() {
```

```
    @Override
```

```
    public void onClick(View view) {
```

```
        // Initializing the popup menu and giving the reference as current context
```

```
        PopupMenu popupMenu = new PopupMenu(MainActivity.this, button);
```

```
        // Inflating popup menu from popup_menu.xml file
```

```
        popupMenu.getMenuInflater().inflate(R.menu.popupmenu, popupMenu.getMenu());
```

```
        popupMenu.setOnMenuItemClickListener(new PopupMenu.OnMenuItemClickListener() {
```

```
            @Override
```

```
            public boolean onMenuItemClick(MenuItem menuItem) {
```

```
                // Toast message on menu item clicked
```

```
                Toast.makeText(MainActivity.this, "You Clicked " + menuItem.getTitle(),
```

```
                    Toast.LENGTH_SHORT).show();
```

```
                return true;
```

```
            } });
```

```
        // Showing the popup menu
```

```
        popupMenu.show();
```

```
    }
```

```
<?xml version="1.0" encoding="utf-8"?>
```

popupmenu.xml

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">
```

```
<item
```

```
    android:id = "@+id/android"
```

```
    android:title = "Android" />
```

```
<item
```

```
    android:id = "@+id/java"
```

```
    android:title = "JAVA"/>
```

```
<item
```

```
    android:id = "@+id/kotlin"
```

```
    android:title = "Kotlin"/>
```

```
</menu>
```

